

D. Destiny

time limit per test: 2.5 seconds
memory limit per test: 512 megabytes

Once, Leha found in the left pocket an array consisting of n integers, and in the right pocket q queries of the form $l\ r\ k$. If there are queries, then they must be answered. Answer for the query is minimal x such that x occurs in the interval $l\ r$ strictly more than $\frac{r-l+1}{k}$ times or -1 if there is no such number. Help Leha with such a difficult task.

Input

First line of input data contains two integers n and q ($1 \leq n, q \leq 3 \cdot 10^5$) — number of elements in the array and number of queries respectively.

Next line contains n integers $a_1, a_2, ..., a_n$ ($1 \leq a_i \leq n$) — Leha's array.

Each of next q lines contains three integers l, r and k ($1 \leq l \leq r \leq n, 2 \leq k \leq 5$) — description of the queries.

Output

Output answer for each query in new line.

Examples

input	Copy
4 2 1 1 2 2 1 3 2 1 4 2	
output	Copy
1 -1	

input	Copy
5 3 1 2 1 3 2 2 5 3 1 2 3 5 5 2	
output	Copy
2 1 2	

Codeforces Round 429 (Div. 1)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
363972293	Feb/23/2026 03:37	Accepted
363971837	Feb/23/2026 03:23	Accepted
363942300	Feb/22/2026 04:00	Accepted
363942140	Feb/22/2026 03:56	Wrong answer on test 2
363942112	Feb/22/2026 03:55	Wrong answer on test 1
363754685	Feb/21/2026 01:48	Accepted
363754194	Feb/21/2026 01:35	Accepted
363754086	Feb/21/2026 01:32	Time limit exceeded on test 6
363753589	Feb/21/2026 01:20	Wrong answer on test 26
363753567	Feb/21/2026 01:20	Time limit exceeded on test 6

→ Problem tags

data structuresprobabilities*2500

No tag edit access

→ Contest materials

Codeforces Round #429

Editorial (en)